



CS & IT ENGINEERING



COMPILER DESIGN

PARSING

Lecture No.- 07



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Recap of Previous Lecture



Topic

??

CLR(1) Parsing

LR(1) item DFA

Closure()

goto()



Topics to be Covered



Topic

??

Topic

??

Topic

??

Parsing



Topic : LALR(1) Parser

① LR(1) items DFA

② If any two (or) more state having same LR(0) part and different lookahead part then merge them into single state and combine all Lookahead symbols.



Topic : CLR(1) Grammar Detection

- shift /reduce conflict (or)

①

$$\begin{array}{l} A \rightarrow \alpha \cdot a B, \{t_1\} \\ B \rightarrow \gamma \cdot \end{array}$$

Im

terminal

$\{t_2\}$

② If $a \in \{t_2\}$ then
S/R Conflict

reduce/reduce conflict
then give grammar is not CLR(1) grammar

①

$$\begin{array}{l} A \rightarrow \alpha \cdot \{t_1\} \\ B \rightarrow \gamma \cdot \{t_2\} \end{array}$$

② $\{t_1\} \cap \{t_2\} \neq \emptyset$ then S/R Conflict

(1)

$S' \rightarrow \cdot S, \$$
 $S \rightarrow \cdot Aa, \$$
 $S \rightarrow \cdot dc, \$$
 $A \rightarrow \cdot d, a$

$S \rightarrow S \cdot, \$$

$A \rightarrow S \cdot a, \$ \xrightarrow{a} S \rightarrow Aa \cdot, \$$

CLR(1) Grammar or not?

$S \rightarrow d \cdot c \{ \$ \}$
 $A \rightarrow d \cdot, \{ a \}$

CLR(1) grammar

② X

No s/r Conflict

②

$S \rightarrow \cdot S, \$$
 $S \rightarrow \cdot AA, \$$

$A \rightarrow \cdot aA, \{a\}$
 $A \rightarrow \cdot a, \{a\}$

a

$A \rightarrow a \cdot \underline{A}, \{a\}$

$A \rightarrow a \cdot, \{a\}$

$(S) \{ A \rightarrow \cdot aA, \{a\} \}$
 $(S) \{ A \rightarrow \cdot a, \{a\} \}$

① ✓

② ✓

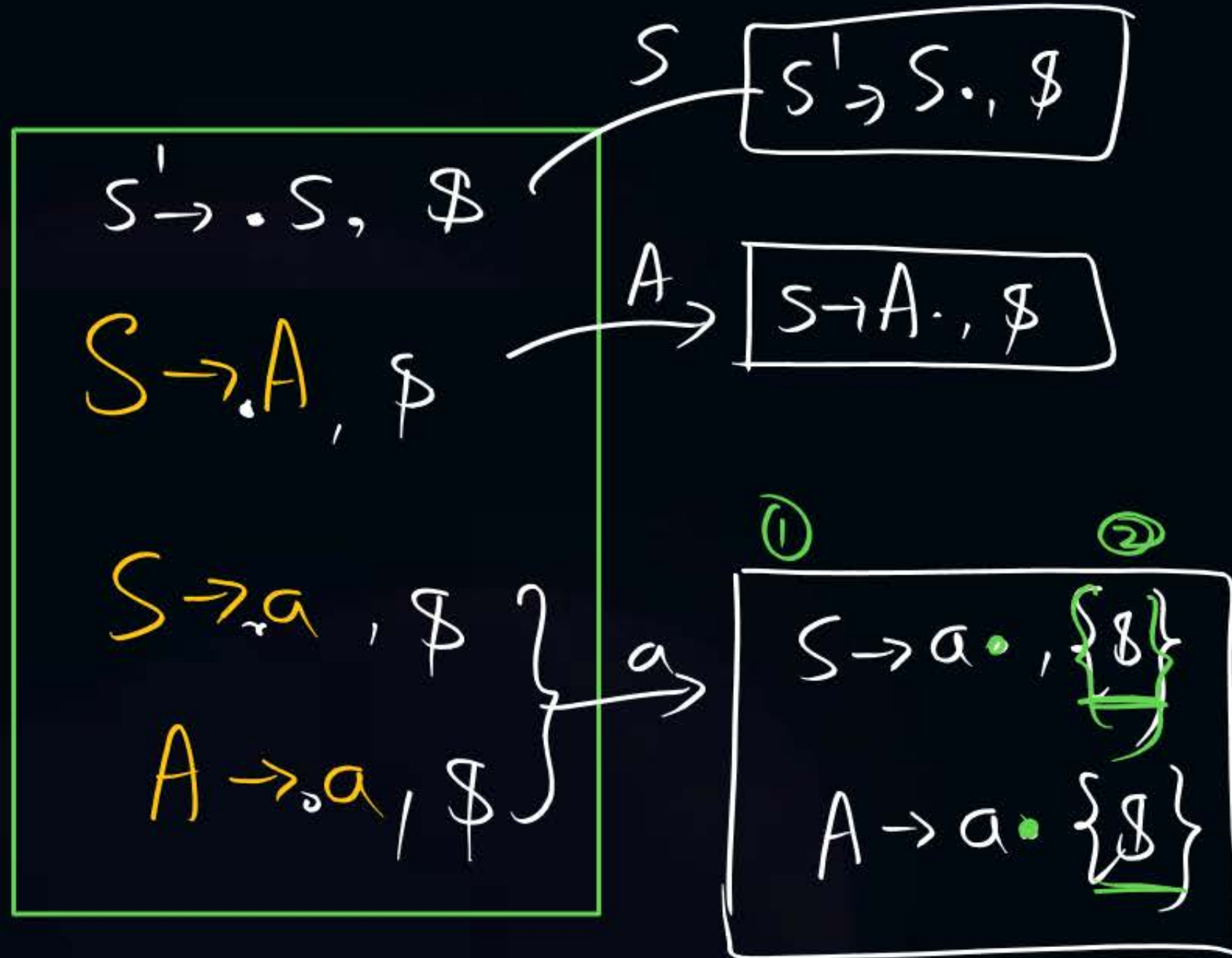
S/R Conflict

CLR(1) grammar not?

NOT CLR(1)

Grammar

①

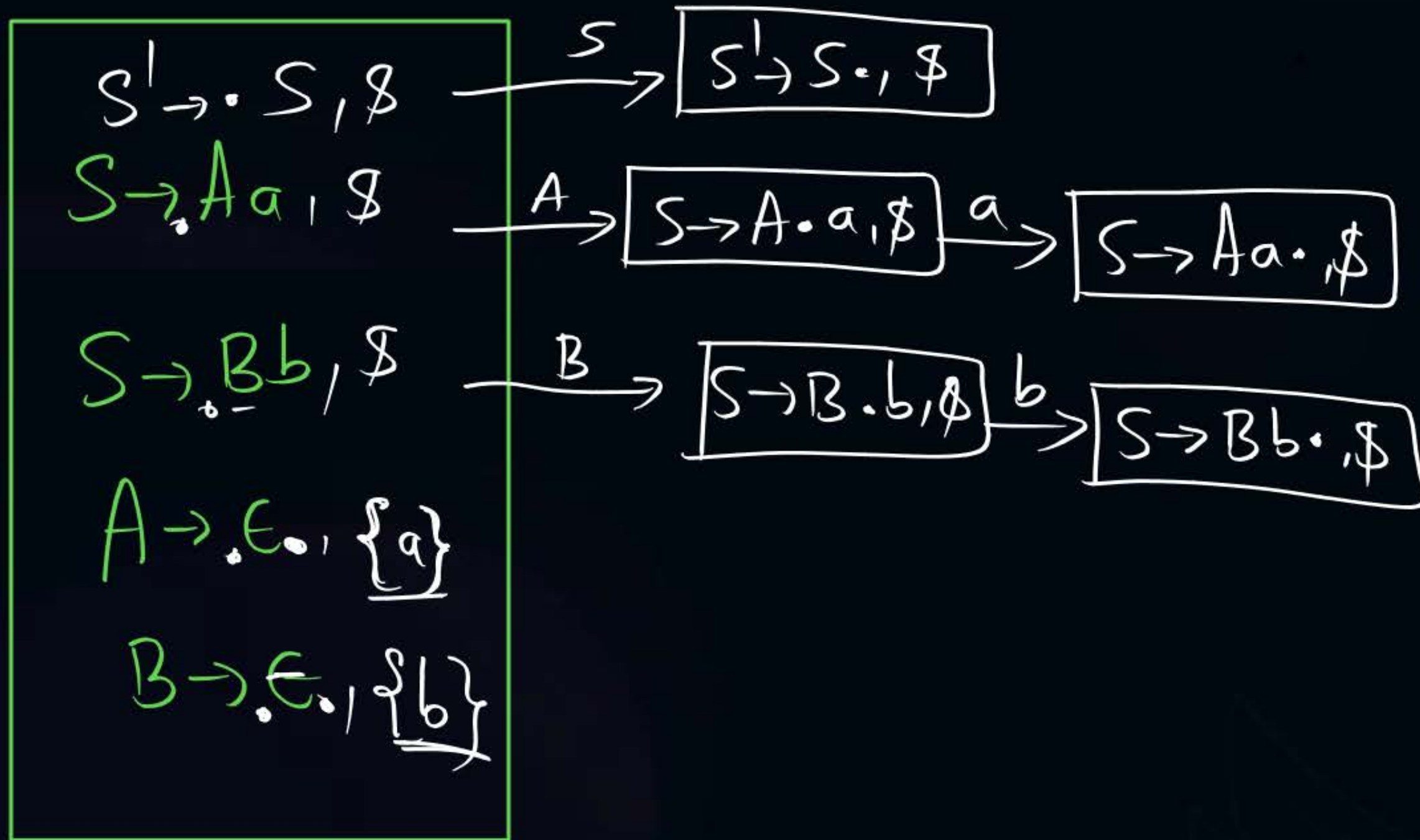


R/R Conflict

CLR(1) grammar
(or) not?

Not CLR(1) Grammar

②



No R/R Conflict

CLR(1) Grammar not?

CLR(1) Grammar

SLR(1) Grammar

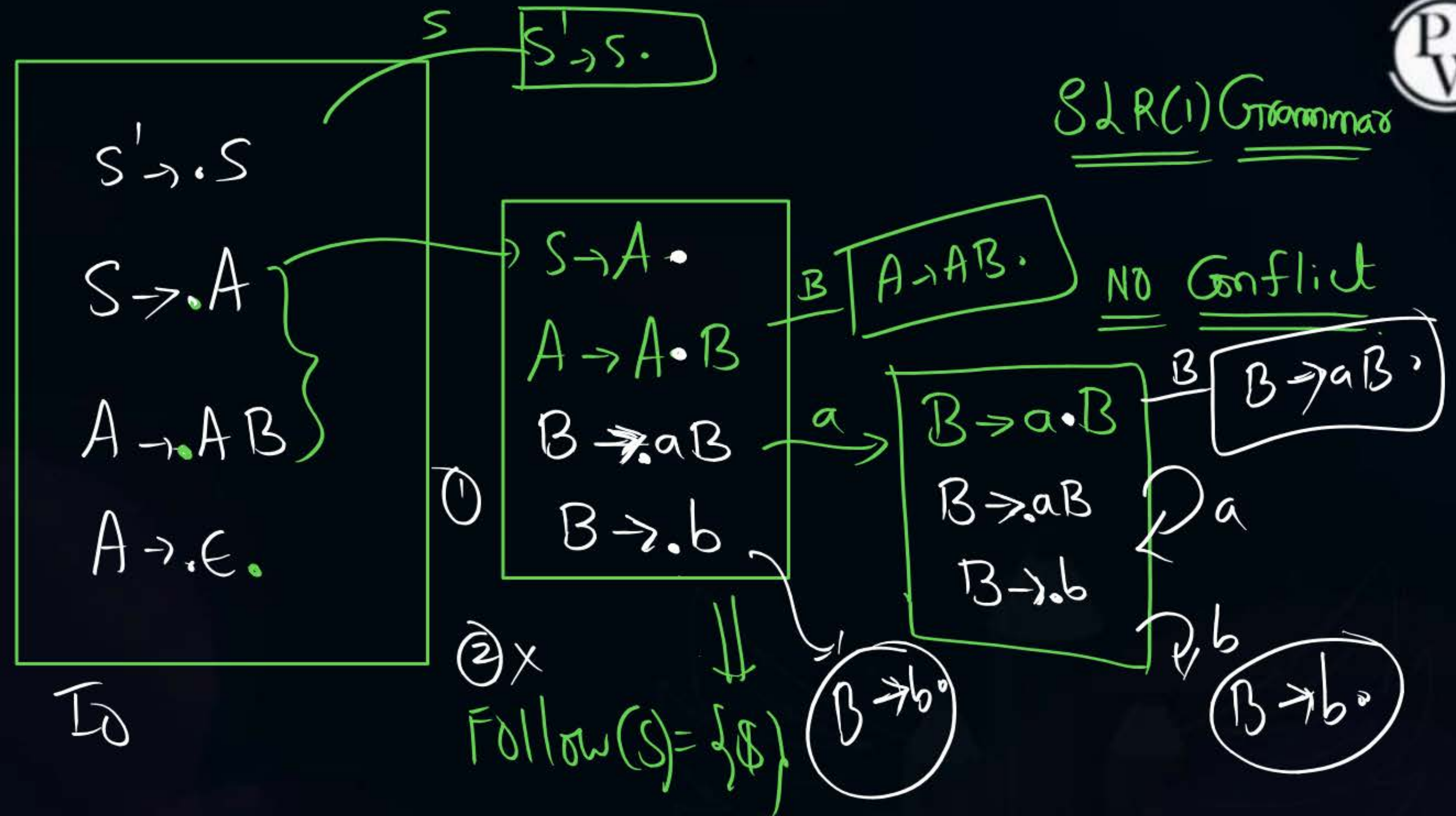
$S \rightarrow A$

$A \rightarrow AB$

$A \rightarrow \epsilon$

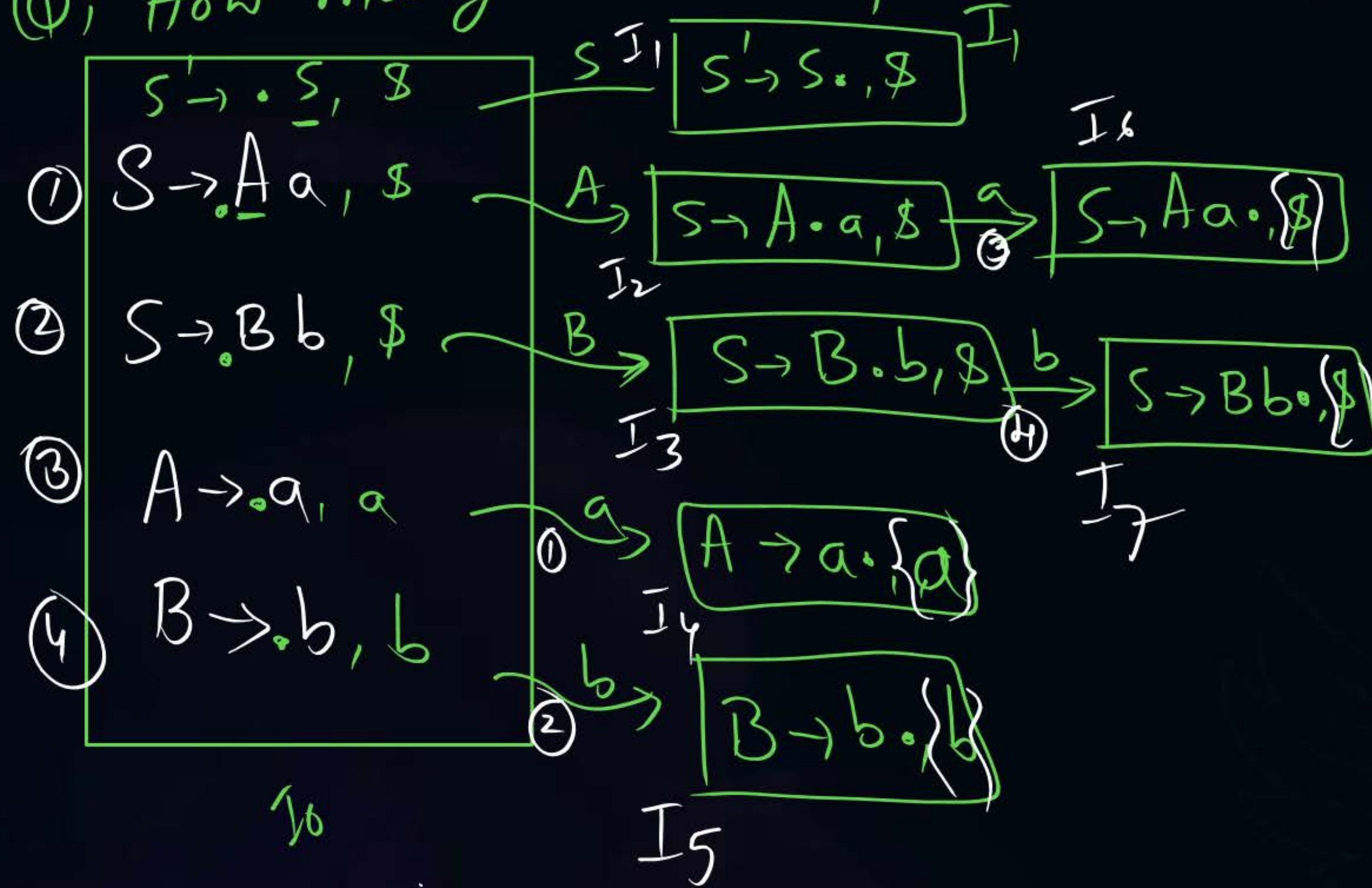
$B \rightarrow aB$

$B \rightarrow b$



No SLR Conflict

① How many states present in LR(1) items DFA for given grammar



① 8 States

② Shift actions = 4 shift

③ GOTO entries = ③

④ Reduce actions

		a	b	\$
δ_3	I_4	δ_3		
δ_4	I_5		δ_4	
δ_1	I_1			δ_1
δ_2	I_7			δ_2



Topic : Conflict States of LALR(1)

- S/R Conflict:

Same of CLR(1)

①

$$\begin{array}{l} A \rightarrow \alpha \cdot a B, \{t_1\} \\ B \rightarrow \gamma \cdot, \{t_2\} \end{array}$$

②

If $a \in \{t_2\}$
then S/R conflict

- R/R Conflict:

①

$$\begin{array}{l} A \rightarrow \alpha \cdot, \{t_1\} \\ B \rightarrow \gamma \cdot, \{t_2\} \end{array}$$

②

$$\{t_1\} \cap \{t_2\} \neq \emptyset$$

then R/R Conflict

#Q. Which of the following grammar is SLR(1) grammar?



No. of States of LR Parser

$$CLR(1) \geq LALR(1) = SLR(1) = LR(0)$$



Topic : CLR(1) Parser & LALR(1)

S/R Conflict

- ①
- (Shift) $A \rightarrow \alpha \cdot a \beta, \{x\}$
- (Reduce) $B \rightarrow \gamma \cdot, \{t_1\}$
- ② if a $\in$ $\{t_1\}$ $\Rightarrow$ S/R Conflict
- terminal*



Topic : CLR(1) Parser

Reduce | Reduce Conflict

I_m

①

$$\begin{array}{l} (R) A \rightarrow \alpha \cdot, \underline{\{t_1\}} \{a, \underline{b}\} \\ (R) B \rightarrow \gamma \cdot, \{ \underline{t_2} \} \{c, d\} \end{array}$$

② if $\{t_1\} \cap \{t_2\} \neq \emptyset$ then Conflict

CLR(1)

<u>I_m</u>	
$A \rightarrow \alpha \cdot$	$\{a\}$
$B \rightarrow \gamma \cdot$	$\{b\}$

<u>I_n</u>	
$A \rightarrow \alpha \cdot$	$\{b\}$
$B \rightarrow \gamma \cdot$	$\{a\}$

CLR(1)

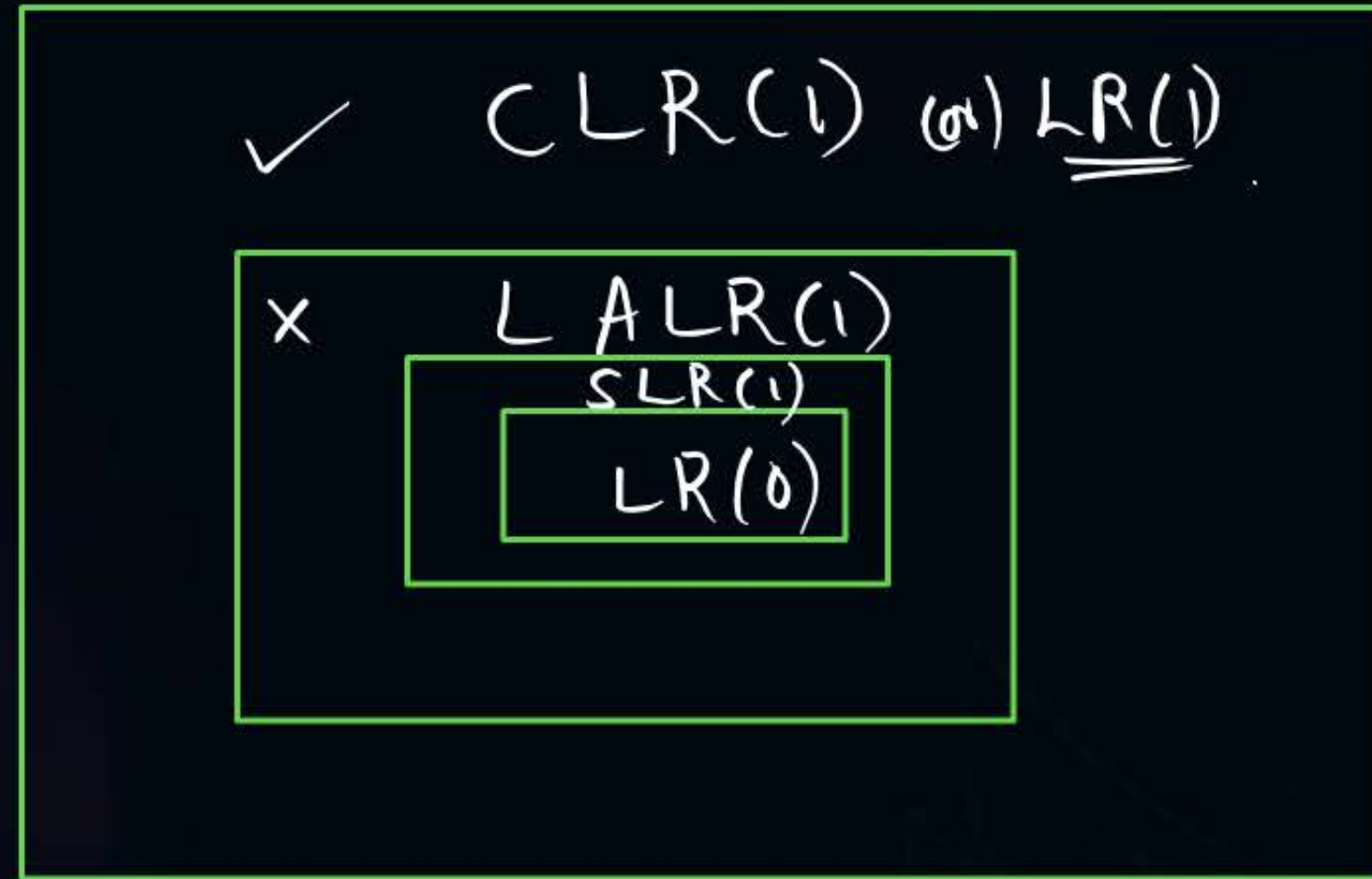
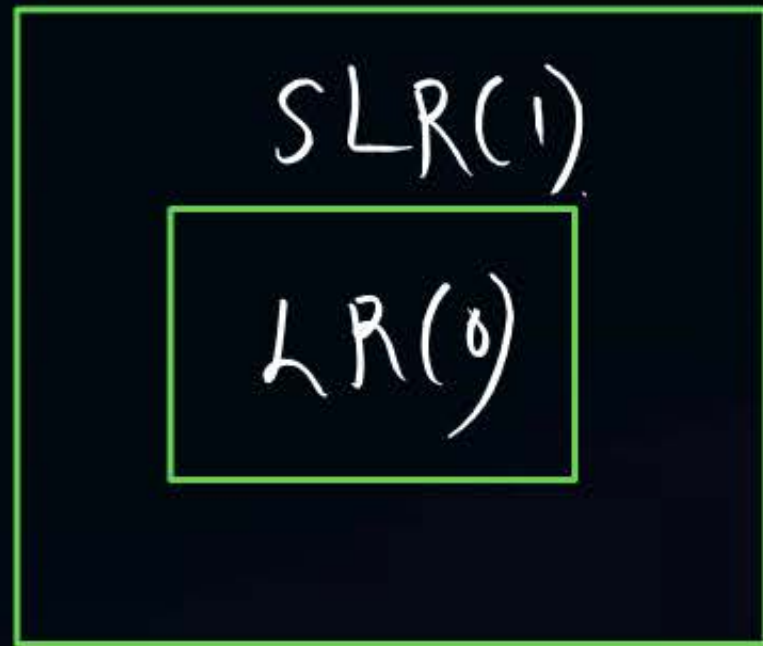
{No R/R
Conflict}

LALR(1)

$A \rightarrow \alpha \cdot$	$\{a, b\}$
$B \rightarrow \gamma \cdot$	$\{a, b\}$

LALR(1)

can have
R/R Conflict



Power of LR Parser

$$CLR(1) > LALR(1) > SLR(1) > LR(0)$$

① Verify Given Grammar is

① CLR(1) Grammar (or) not

② LALR(1) Grammar (or) not

② Find Number of States of
CLR(1) & LALR(1) Parser

$S \rightarrow Aa$

$S \rightarrow bAc$

$S \rightarrow Bc$

$S \rightarrow bBa$

$A \rightarrow d$

$B \rightarrow d$

Home Work



① Verify Given Grammar is

① CLR(1) Grammar (or) not

② LALR(1) Grammar (or) not

② Find Number of States of
CLR(1) & LALR(1) Parser

$S \rightarrow Aa$
 $S \rightarrow bAc$
 $S \rightarrow dc$
 $S \rightarrow bda$
 $A \rightarrow d$

① If No S/R Conflict In CLR(i)
then

S/R Conflict will not prevent LALR(i) also



2 mins Summary



Topic

One

CLR(1) ✓

Topic

Two

LALR(1) ✓

Topic

Three

Topic

Four

Topic

Five



THANK - YOU

CLR(1) = 10
SLR(1) = 6

Drawback of LR Parser

$S \rightarrow (S)$

$S \rightarrow a$

Number of states of CLR Parser is
more compared to other Parsers.

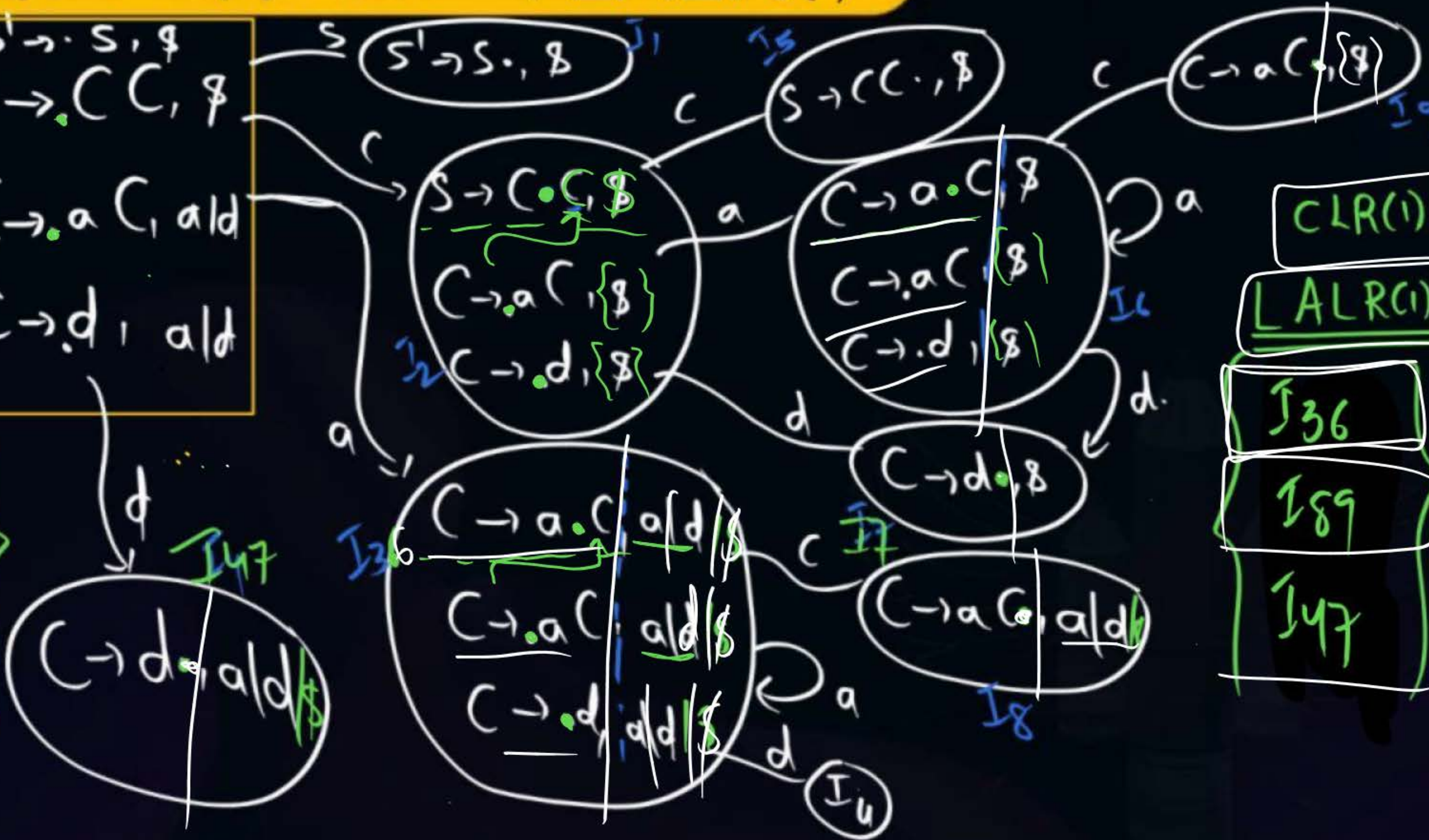
Hence LALR(1) Parsing Tables constructed



Topic : CLR(1) Parser & LALR(1)

$S' \rightarrow \cdot S, \$$
 $S \rightarrow \cdot C C, \$$
 $C \rightarrow \cdot a C, a | d$
 $C \rightarrow \cdot d, a | d$

$\{ S \rightarrow CC \}$
 $\{ C \rightarrow aC \}$
 $\{ C \rightarrow d \}$



CLR(1): 10 states

LALR(1) = 7 states

I36
I89
I47

Number of states of LR Parser

Number of states of

$$CLR(1) \geq LALR(1) = SLR(1) = LR(0)$$

$\Downarrow$
10

$\Downarrow$
6

$\Downarrow$
6

$\Downarrow$
6

① GATE
 $\left\{ \begin{array}{l} S \rightarrow (S) \\ S \rightarrow a \end{array} \right\} \Rightarrow$

② $\left\{ \begin{array}{l} S \rightarrow CC \\ C \rightarrow a \\ C \rightarrow aC \end{array} \right\} \Rightarrow$

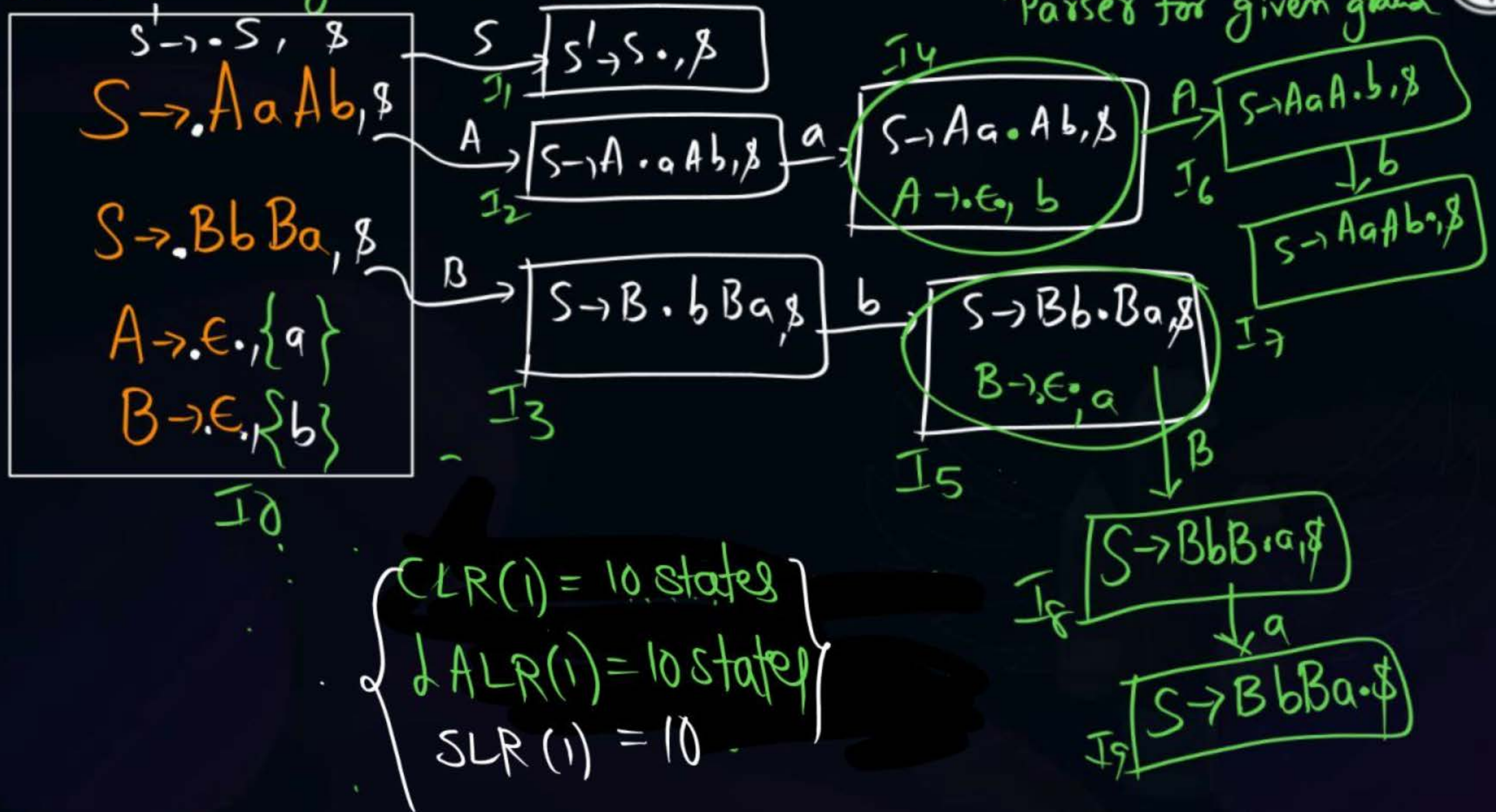
10

7

7

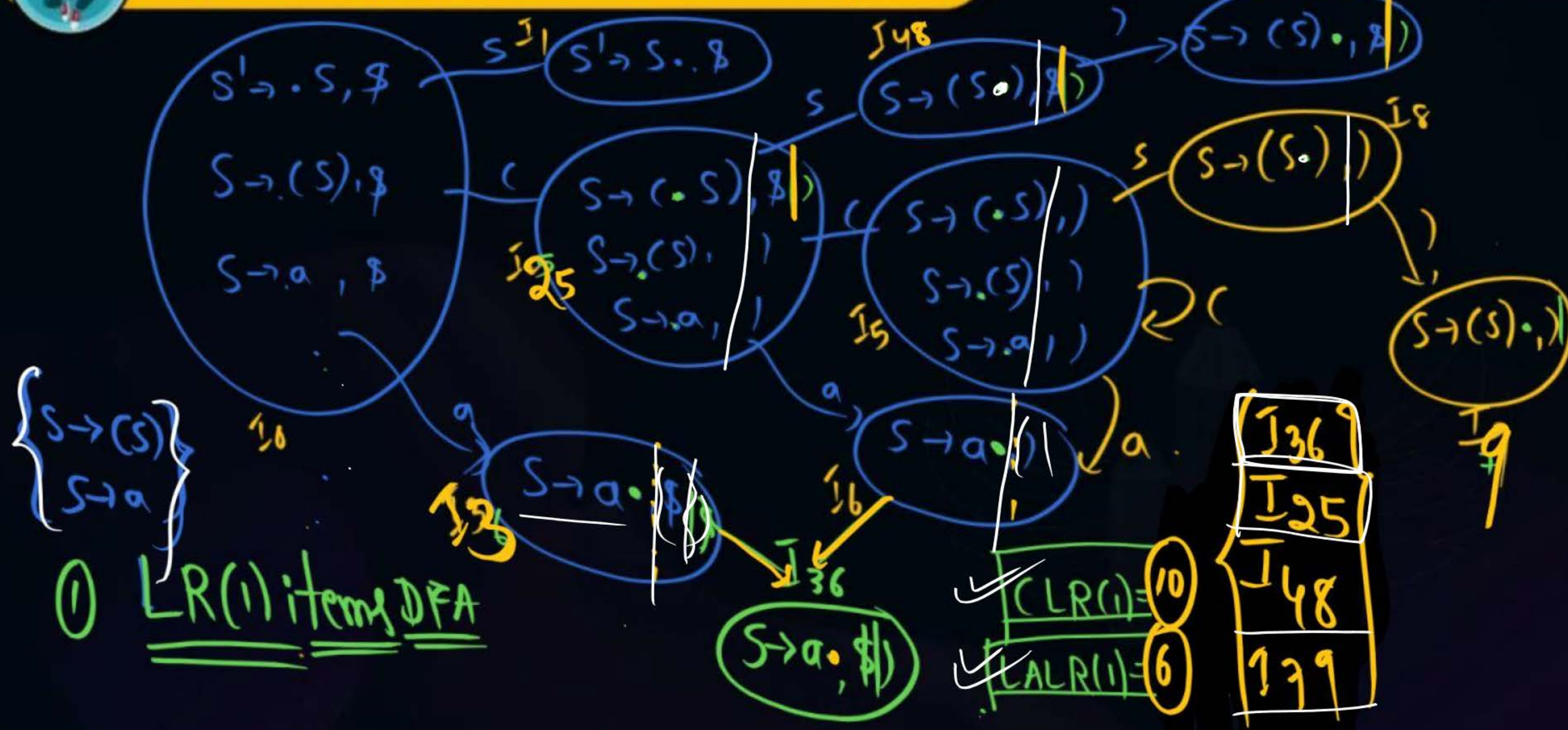
7

(Q) How many states present in CLR(1), LALR(1) and SLR(1) Parser for given grammar





Topic : CLR(1) Parser & LALR(1)



① LR(1) item DFA

✓ CLR(1) = 10
✓ LALR(1) = 6

I ₃₆
I ₂₅
I ₄₈
I ₉